

Module TH-37: The D·A·M taxonomy

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After this module you can:

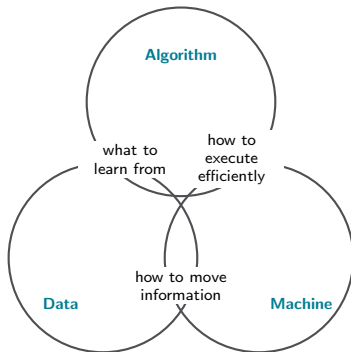
- ▶ Name the three axes of the D·A·M taxonomy and say what each axis controls.
- ▶ Read a slow system as three terms of one equation, and say which term the measurement points at.
- ▶ Name the bottleneck of a failing system before anyone optimizes it.

A device misses its deadline. The number is on the dashboard, and the team has one hour to act.

- ▶ Three groups are in the room. One owns the sensors, one wrote the model, and one owns the chip.
- ▶ Each group has an improvement ready. None of them has a measurement.

Key point: Optimizing before measuring picks a random axis and hopes.

Three axes, one system



A change on one axis moves the constraint of the other two. That is why one number never tells the whole story.

Three words appear in the rest of this module. An **accelerator** is a separate chip that does the arithmetic of a model. A **FLOP** is one floating-point operation. A **byte** is one unit of data.

What each axis controls

Axis	Its limit	Its high-leverage change
Data Algorithm	Volume, quality, arrival rate Operations and dependencies	Take less data, or better data Compress the model
Machine	Compute, memory, I/O	Change the hardware

Work on the Data axis alone and the load moves. Less data and cleaner labels raise the quality of the information, and they also cut the number of bytes that the Machine axis has to move.

The interesting places are the overlaps

Zone	The question it asks
Data alone	Is the information itself right?
Algorithm alone	Is the logic right?
Machine alone	Is the physics fast enough?
Data and Algorithm	What can the model learn from this data?
Data and Machine	Can the machine be fed fast enough?
Algorithm and Machine	Does the code execute this logic well?
All three at once	Is this a system, and is it measured?

A single-axis problem is rare. Most slow systems sit on a boundary, and a change on one side of a boundary moves the load to the other side.

$$T = \underbrace{\frac{D_{\text{vol}}}{\text{BW}}}_{\text{data term}} + \underbrace{\frac{O}{R_{\text{peak}} \cdot \eta_{\text{hw}}}}_{\text{compute term}} + \underbrace{L_{\text{lat}}}_{\text{latency term}}$$

- ▶ The data term counts the bytes and the bandwidth that move them.
- ▶ The compute term counts the operations, the peak rate of the hardware, and how much of that rate the run reaches.
- ▶ The latency term is the fixed cost of launching, waiting, and synchronizing.

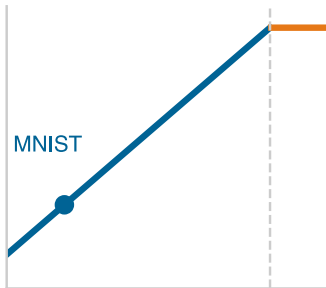
The three terms are the three axes in units of seconds. The largest one is the part that costs the time.

The largest term names the axis

If the run is	Then fix	Do not bother with
Memory bound	Quantize, prune, fuse, batch	A faster arithmetic unit
Compute bound	Better kernels, lower precision	More bandwidth
Latency bound	Shorter chain, fewer launches	Any extra compute

To **quantize** is to store a number in fewer bits. To **prune** is to drop weights. To **fuse** is to join two operations, so that the data between them moves once. **Batching** sends several inputs at once, so one weight read serves all of them. The last column is the expensive column. A faster accelerator does nothing for a workload that waits for its data.

One number separates memory from compute



- ▶ The arithmetic intensity of a workload is its operations per byte moved.
- ▶ The ridge point of a chip is its peak rate divided by its bandwidth. For a server accelerator it sits between 150 and 300 FLOP per byte.
- ▶ Below the ridge the workload is memory bound. Above it, the workload is compute bound.

The dot on the slope is a small MNIST digit classifier, and it needs about 12 FLOP for each byte it moves. That is more than ten times below the ridge point, so it stays on the slope even on a large accelerator.

Source: Figure adopted from Volume I, chapter 5.

Read the dashboard before you touch the code

The reading	What it usually means
Accelerator under 80 percent busy	Data, host, or launch starvation
Accelerator over 95 percent busy	Compute bound, unless the memory bus is also full
Batch size of one	Latency, not throughput, is the goal
Data wait above 10 percent of a step	The pipeline, not the model
Model FLOP utilization under 30 percent	The work of the model itself is small next to the work of the run

Batch size is the number of inputs the model handles at one time. **Model FLOP utilization** is the rate the model reaches, divided by the peak rate of the accelerator.

These are screens, not proofs. Each one names the measurement that confirms it.

Three ways to optimize the wrong axis

- ▶ **The hardware crutch.** A faster accelerator is bought for a slow data loader. The new chip waits faster.
- ▶ **The model twiddle.** The architecture is changed while the bottleneck sits in the disk or in the network.
- ▶ **The premature optimizer.** A custom kernel is written while the model still repeats the same operation twice.

A **kernel** is the compiled code that runs one operation on the accelerator.

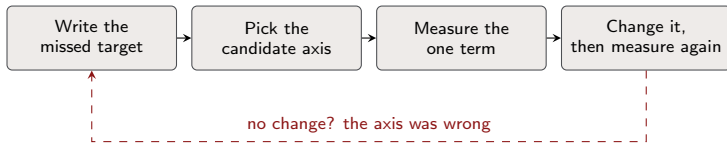
The three traps have one cause: acting before the measurement.

Three cases, three axes

Symptom	Axis	The move that follows
Accelerator between 10 and 40 percent, and gaps while it waits	Data	Decode each input once, and send the result early
20 ms budget missed at batch size one	Algorithm and Machine	Shorten the chain, or quantize the weights
Accelerator at 99 percent, memory bus idle	Machine	Lower the precision, or add a bigger chip

The same reading of the dashboard sends two of these cases to different teams.

The method in four passes



The loop is the method. The label of the axis is a hypothesis, and the last pass is the only one that settles it.

1. Three axes: Data, Algorithm, and Machine. A change on one moves the constraint of the other two.
2. Time is three terms: the bytes over the bandwidth, the operations over the peak rate, and the fixed latency. The largest term names the axis.
3. The last column of the diagnostic table is the expensive one.
4. The axis is a hypothesis. The measurement settles it.

Key point: Name the bottleneck before you optimize it.

Exercise: Name the bottleneck

A team has three slow systems.

1. A speech service on a fast accelerator answers in 340 ms, and the accelerator is idle 60 percent of the time. A log of the run shows the gaps, and the audio files are decoded during them.
2. A camera on a microcontroller runs the same network in 95 ms. The same network runs in 12 ms on a laptop, and the two use the same number of operations.
3. A training job uses a new accelerator with twice the arithmetic of the old one. The step time falls by 4 percent.

For each system, name the bottleneck and give the one measurement that would confirm it.

1. **Data.** The accelerator is idle because it waits for decoded audio. The confirming measurement is the data wait time against the step time. Adding arithmetic changes nothing.
2. **The Data and Machine boundary.** The same work on a different machine finishes eight times faster, so the arithmetic was not the limit: the run was memory bound. The confirming measurement is the arithmetic intensity against the ridge point of each machine.
3. **The Data and Machine boundary.** Twice the arithmetic gave 4 percent, so the arithmetic was not the binding term. The confirming measurement is the arithmetic intensity of the job against the ridge point of the new accelerator. Below the ridge, the extra arithmetic sat unused.

Two of the three cases point at the data term. That is the usual result, and it is why the first measurement is a wait time and not a speed number.

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.